Quiz 2: Prediction and reward

Read a paper, replicate it, explain it. Sessions 7–12 · 90 minutes · 40 marks · open book

Name _____ Roll no. _____

The paper

Schultz, W., Dayan, P., & Montague, P. R. (1997). A neural substrate of prediction and reward. *Science* 275, 1593–1599. Read it on your device (through your library) or on a printed copy.

https://doi.org/10.1126/science.275.5306.1593

The lab: Prediction Lab (password on the screen)

Runs in any browser. Nothing is saved or sent: **write every number in this booklet**.

rakesh-sengupta.github.io/neural-signalling-labs/labs/prediction-lab

Suggested timing (you may work in any order):

Section	What you do	Marks	Minutes
A Reading	Read the paper with the guide on page 2; answer on page 3	10	20
B Replicating	Run the four parts of Prediction Lab; fill in pages 4–6	15	35
C Connecting	Link the paper to Sessions 7–12 (pages 7–8)	10	25
D Thinking	One short argued answer, from a choice of three (page 9)	5	10

Rules

Open book: you may use the paper, your booklets, notes and the course slides.

Your device is for the paper and the lab only: no messaging, no AI tools, no search.

Work alone. If the lab will not load, raise your hand.

Short answers are fine. A clear sentence beats a long paragraph. Marks are shown as [2].

Reading guide

The paper in one sentence. Dopamine neurons do not simply fire when a reward arrives: they fire when a reward is *better than predicted*, stay quiet when it is *as predicted*, and pause when it is *worse than predicted*, exactly like the error signal of a learning rule from computer science.

What to read, in this order (about 15 minutes):

Page 1593	The abstract and the first two columns: what "prediction" and "reward" mean; blocking.
Page 1594	**Read closely:** "Information encoded in dopaminergic activity" and **Fig. 1** with its legend.
Page 1595	Skim. The equations are the learning rule; you do *not* need to work with them. Read Eq. 3 in words: *error = reward now + prediction of what comes next – prediction a moment ago*.
Page 1596	**Read closely:** the first column (the model's two testable predictions) and **Fig. 3** with its legend. Skim the maze example.
Pages 1597–1598	Skim "Summary and future questions". Note 18 (page 1598, references) matters for Section D.

Words you will meet

dopamine neuron	a neuron that releases dopamine; these are in the midbrain (VTA and substantia nigra) and send axons to the striatum, nucleus accumbens and frontal cortex
CS (conditioned stimulus)	a cue that has come to predict a reward, such as the light
US / primary reward	the reward itself, here a drop of fruit juice
phasic	brief: a burst or a pause lasting a fraction of a second
basal firing rate	the neuron's resting rate of spikes when nothing is happening
raster	a plot with one row per trial and one dot per spike
peri-event time histogram	spikes counted in small time bins and added across trials, lined up on an event
prediction error	reward received minus reward predicted: positive, zero or negative
TD (temporal difference)	a learning rule that updates predictions from one moment to the next
blocking	when an already-predicted reward stops a new cue being learned

Section A: Reading the paper

10 marks · about 20 minutes

A1. Look at **Fig. 1**. In the lower part of each panel, what is one *row*, and what is one *dot*? What does the histogram above each raster show? **[2]**

A2. Fill in the table for the three panels of Fig. 1. **[3]**

Panel	Was the reward predicted? Did it come?	What the neuron does at the time of the reward	Prediction error (+, 0, –)
Top			
Middle			
Bottom			

A3. The legend says the dip in the bottom panel happens "more than 1 s after the conditioned stimulus without any intervening stimuli". Why is that detail important? What does it show about the brain? **[2]**

A4. Before anyone recorded single dopamine neurons, three kinds of evidence linked dopamine to reward (page 1594). Name all three, in a few words each. **[3]**

1. _____

2. _____

3. _____

Section B: Replicating the paper

15 marks · about 35 minutes

Open Prediction Lab (page 1) and follow each part's steps. **Your numbers will differ a little from your neighbour's**: real recordings vary too.

B1. Be the monkey Lab Part 1, about 4 minutes

Time from star to press, trials 1–4		ms
Time from star to press, last 6 ordinary trials		ms
Time from star to press, the 2 early-star trials		ms
No-star trials: how many you pressed in, and when (after the light)	of 3, at	ms

(a) Compare your first two numbers. What had you learned? **[1]**

(b) Were you slower or faster when the star came 0.5 s early? Why? **[1]**

(c) On the no-star trials, what did you do, and when? Which panel of Fig. 1 is this like, and why? **[1]**

B2. Record a dopamine neuron Lab Part 2, about 8 minutes

Recording (spikes per second)	**Resting**	**After the light**	**Juice time**
Untrained: light then juice			
Trained: light then juice			
Trained: light, juice withheld			
Trained: juice, no light			

[1]

(a) Which of your recordings matches each panel of Fig. 1? **[1]**

Top: _____ Middle: _____ Bottom: _____

(b) After training, the neuron still bursts for juice that comes *without* the light. Why, according to the theory? **[1]**

(c) Did the replication work? Name one way your model recording differs from the real neuron in Fig. 1. **[1]**

B3. Watch it learn Lab Part 3, about 10 minutes

Use the slider to move through the 80 trials. The readout gives the prediction error at the light and at juice time.

Run	First trial with the juice response below 0.5	First trial with the biggest response at the light	Response at the light, trial 80
One light			
One light + receptor blocker			
Two lights			light 1: light 2:

[1]

(a) The legend of Fig. 3 says “the timing of reward delivery is learned well before any response transfers to the earliest sensory cue”. Do your numbers agree? Quote them. **[1]**

(b) What happened on trial 50, and where do you see it in Fig. 3 of the paper? **[1]**

(c) With two lights, which light ends up with the response? Which of the model’s two “testable predictions” (page 1596) does this check? **[1]**

B4. Test the theory Lab Part 4, about 12 minutes

The neuron is fully trained: light at 0 s, juice at 1.0 s. **Predict first**: mark each cell **+** (burst), **0** (no change) or **–** (pause) *before* you record. Then write the rate the lab shows.

Test prediction / rate	**at 0 s**	**at 0.5 s**	**at 1.0 s**	**at 1.5 s**
A. As trained	/	/	/	/
B. Juice withheld	/	/	/	/
C. Juice 0.5 s late	/	/	/	/
D. Juice 0.5 s early	/	/	/	/
E. Double the juice	/	/	/	/
F. Half the juice	/	/	/	/
G. Light 2 left out	/	/	/	/

[2]

B4 (a) In Test F the monkey still gets juice, yet the neuron pauses. Explain this using "prediction error". **[1]**

B4 (b) Test G checks the model's second testable prediction (page 1596). What did the neuron do at 0.5 s and at 1.0 s? Explain both. **[1]**

Before you move on: check Section B

☐ B1 table and (a)–(c) ☐ B2 table and (a)–(c) ☐ B3 table and (a)–(c) ☐ B4 table and (a)–(b)

Keep the lab open: Sections C and D may send you back to it.

Section C: Connecting to Sessions 7–12

10 marks · about 25 minutes

C1. Spikes (Session 9). Every spike is all-or-none. So how can one neuron say both "better than expected" and "worse than expected"? Real dopamine neurons rest at only about 0.5–7 spikes per second. Why does that make "worse than expected" harder to signal? **[2]**

C2. Synapses (Sessions 11–12). Suggest a synaptic input that could cause the *burst*, and one that could cause the *pause*. For each, name a likely transmitter and what happens at the membrane (which ions, EPSP or IPSP). **[2]**

C3. Conduction (Session 10). Dopamine axons are thin and unmyelinated; in rats they conduct at about 0.5 m/s. Suppose the burst must travel 4 cm to the frontal cortex. How long does that take? How long would a 10 µm myelinated axon take (about 6 m/s per µm)? Show your working. **[2]**

C4. Receptors (Session 12). All dopamine receptors (D1–D5) are G-protein-coupled (Bear, Table 6.2). From Relay Lab Part 3, what does this suggest about how fast, and for how long, dopamine affects its target neurons? The paper ends by saying such systems "may also deliver precisely timed information". Is there a tension? **[2]**

C5. Drugs (Session 12). Cocaine and amphetamine block the reuptake of dopamine. (a) What happens to dopamine in the gap after a burst? (b) Using the theory, suggest why a drug that does this could make an experience seem "better than expected" every time, so it is never fully predicted. **[2]**

Section D: Thinking

5 marks · about 10 minutes

Choose **one**. Write about 150 words. You will be marked on a clear claim, evidence from the paper or your lab, and one honest limitation. **My choice:** ☐ D1 ☐ D2 ☐ D3

D1. A replication that fails

Look at your Test D (juice 0.5 s early). Now read note 18 of the paper (page 1598). Where does the model disagree with the real neurons? Does this mean the theory is wrong, or something narrower? What would you change in the model?

D2. "Dopamine is the brain's pleasure chemical"

Use the paper and your lab results to evaluate this popular claim. Which findings fit it, and which do not?

D3. Who is surprised?

The paper says dopamine neurons "code for" an error and are "feature detectors of the goodness" of events. Is the neuron surprised? Recall the homunculus from Session 9: who, or what, reads this error signal, according to the paper?

End of Quiz 2. Check that your name is on page 1.

Rough work

This page is not marked.